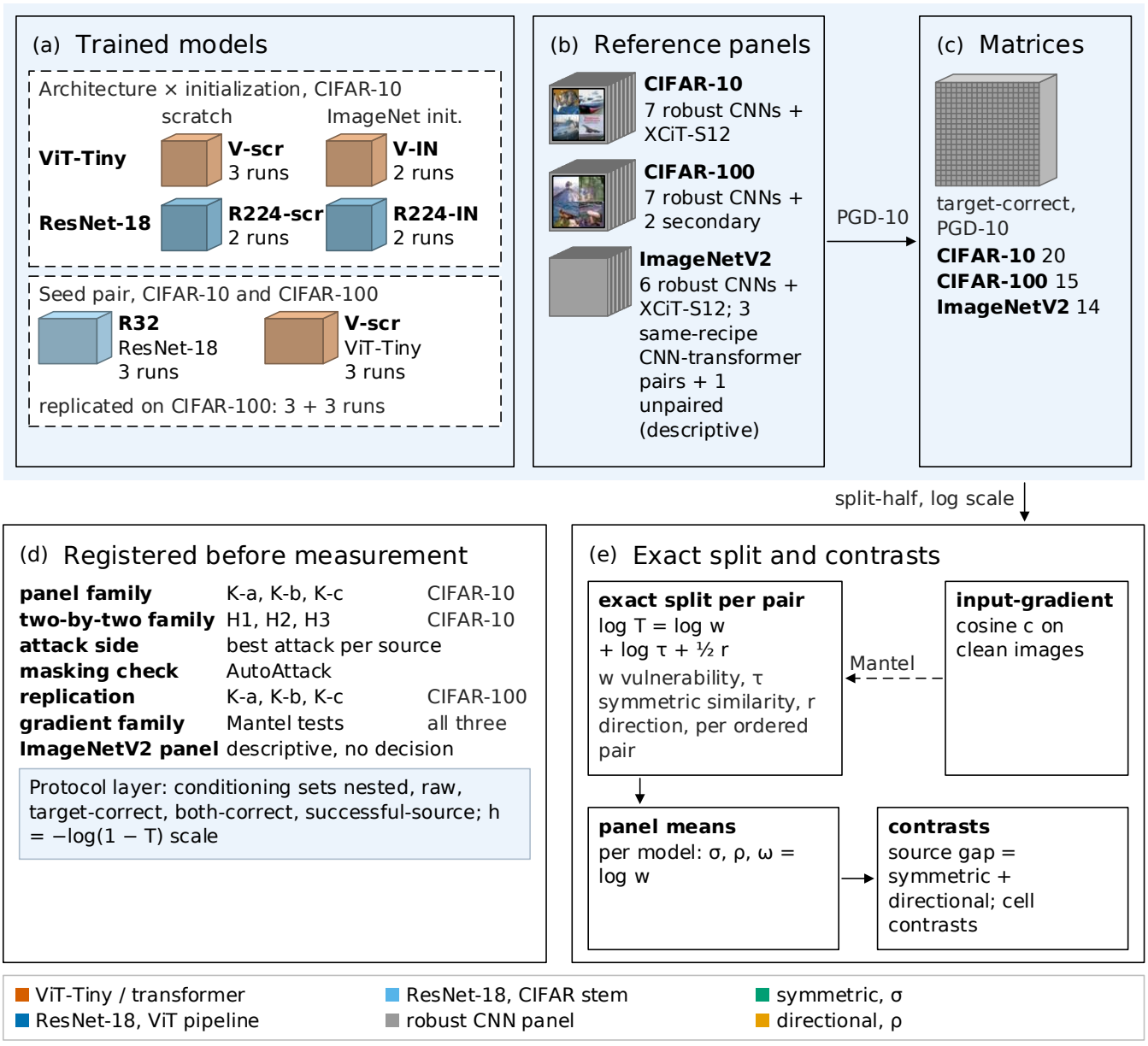

(a) Trained models
Architecture × initialization, CIFAR-10
scratch
ImageNet init.
ViT-Tiny
V-scr
3 runs
V-IN
2 runs
ResNet-18
R224-scr
2 runs
R224-IN
2 runs
Seed pair, CIFAR-10 and CIFAR-100
R32
ResNet-18
3 runs
V-scr
ViT-Tiny
3 runs
replicated on CIFAR-100: 3 + 3 runs
(b) Reference panels
CIFAR-10
7 robust CNNs + XCiT-S12
CIFAR-100
7 robust CNNs + 2 secondary
ImageNetV2
6 robust CNNs + XCiT-S12; 3 same-recipe CNN-transformer pairs + 1 unpaired (descriptive)
PGD-10
(c) Matrices
target-correct, PGD-10
CIFAR-10 20
CIFAR-100 15
ImageNetV2 14
split-half, log scale
(d) Registered before measurement
panel family K-a, K-b, K-c CIFAR-10
two-by-two family H1, H2, H3 CIFAR-10
attack side best attack per source
masking check AutoAttack
replication K-a, K-b, K-c CIFAR-100
gradient family Mantel tests all three
ImageNetV2 panel descriptive, no decision
Protocol layer: conditioning sets nested, raw, target-correct, both-correct, successful-source; h = −log(1 − T) scale
(e) Exact split and contrasts
exact split per pair
log T = log w + log τ + ½ r
w vulnerability, τ symmetric similarity, r direction, per ordered pair
Mantel
input-gradient
cosine c on clean images
panel means
per model: σ, ρ, ω = log w
contrasts
source gap = symmetric + directional; cell contrasts
ViT-Tiny / transformer
ResNet-18, ViT pipeline
ResNet-18, CIFAR stem
robust CNN panel
symmetric, σ
directional, ρ